Explainable AML Transaction Monitoring

Bank operating patterns and a full-data SAML-D model comparison



Class project · October 2026

Executive readout

9.50M	0.104%	0.990
synthetic transactions	positive labels	XGBoost test AP

Selected result

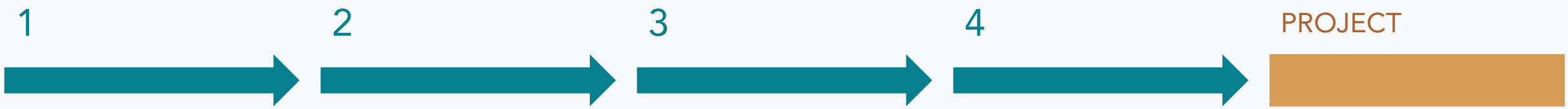
XGBoost scored every test transaction and flagged 1,708. It caught 1,650 of 1,683 synthetic positives with 58 false alerts.

Decision boundary

Track alert unions are diagnostics. The selected queue is XGBoost alone; the 72,795-alert combined queue is not recommended.

Four monitoring patterns and one project example

Illustrative operating patterns, followed by this project's distinct all-transaction MVP.

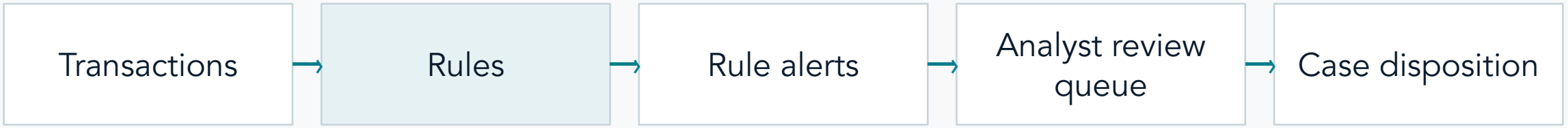


Rules	Risk ranking	Parallel detection	ML alert triage	This MVP
Rules create alerts for analyst review	Ranks rule alerts for review	Rules and a model create alerts	ML ranks consolidated alerts	All transactions scored; selected model creates alerts

The design questions across patterns

What enters the candidate pool? What determines review priority? How is added coverage measured against workload?

Pattern 1: rules and analyst review



Where this works

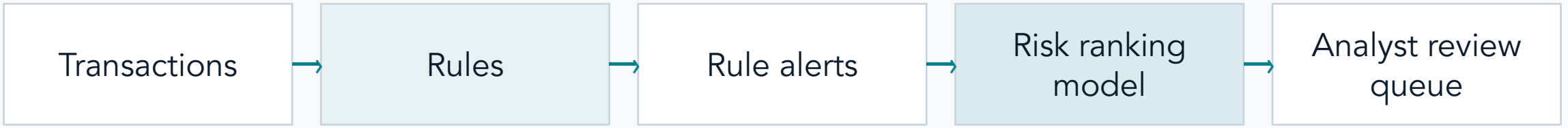
Known typologies translate into readable conditions and repeatable controls.

Where it strains

Review capacity and rule coverage can diverge as transaction volume and behavior change.

Bank example (2006): M&T described rule-selected transactions and analyst review.

Pattern 2: risk ranking of rule alerts



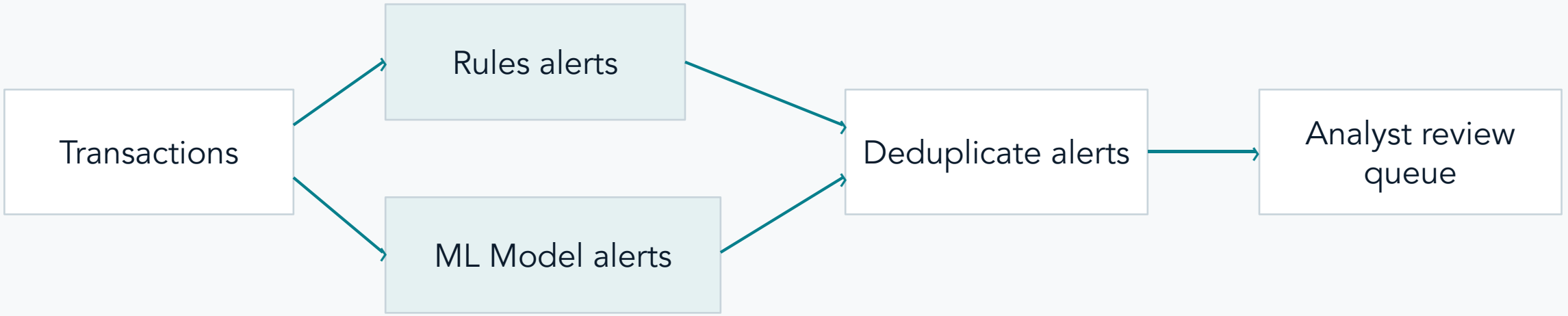
The recall ceiling

Because it scores only rule alerts, the model cannot recover a transaction the rules never surfaced.

For a fixed labeled population, model recall cannot exceed rule recall.

Bank example (2020): Capital One used a random forest to rank AML alerts.

Pattern 3: parallel data science detection

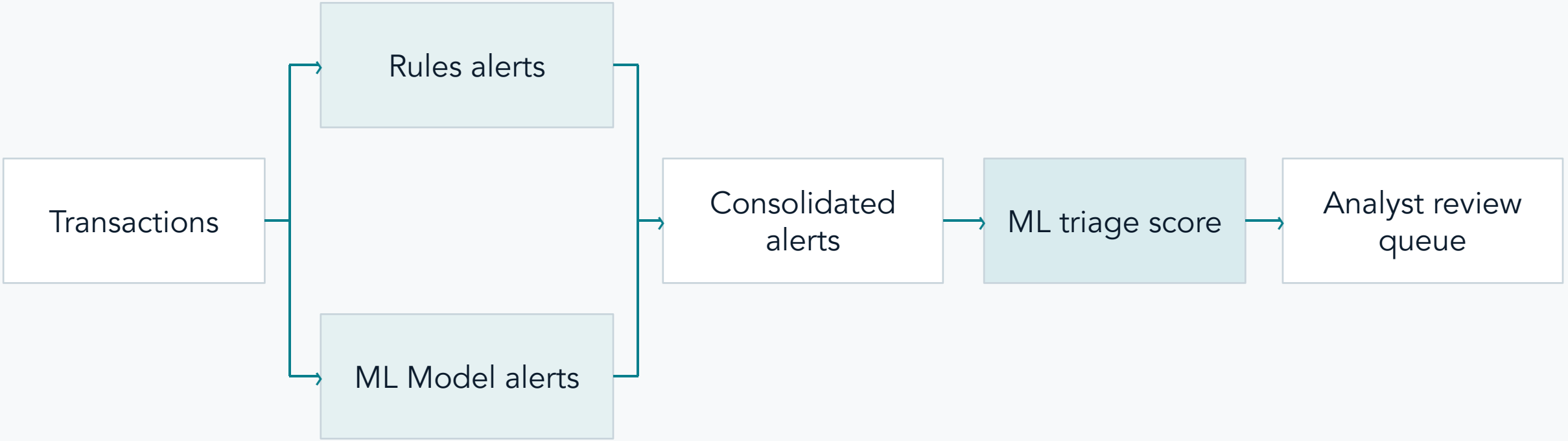


What changes

The DS track can find candidates outside the rules. The bank must measure unique catches and the new workload after deduplication.

Bank examples: ING reported live anomaly detection; OCBC planned a parallel pilot.

Pattern 4: ML triage of consolidated alerts

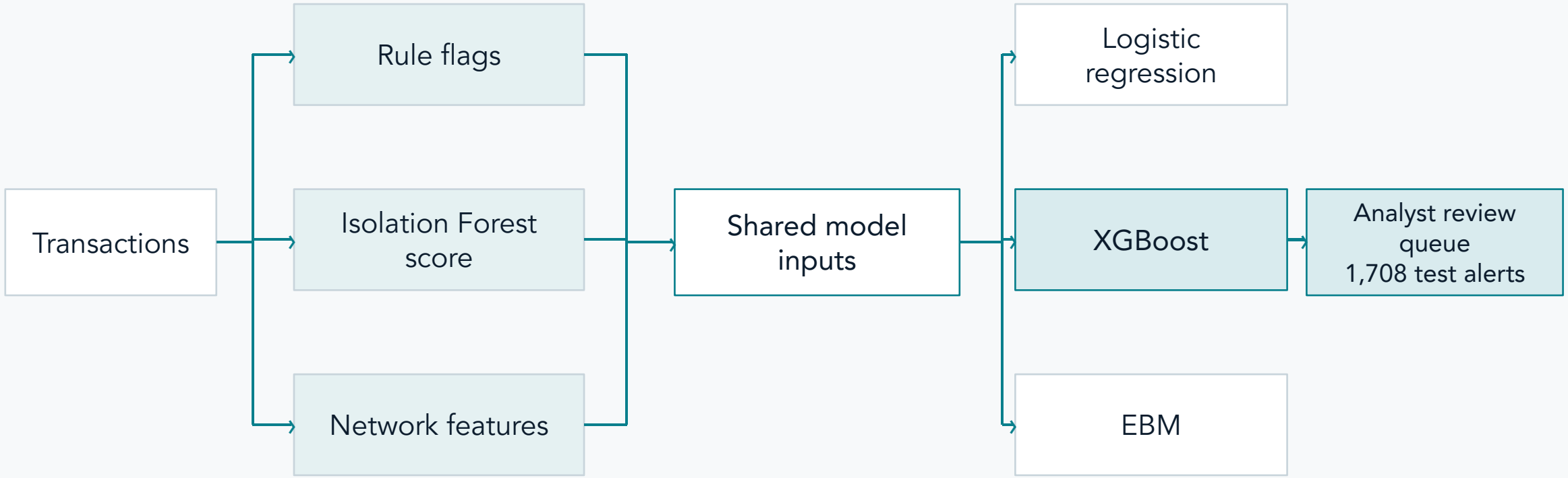


What changes

A separate ML model prioritizes the consolidated alerts for analyst review.

Bank example: UOB reports ML-based prioritization of transaction-monitoring alerts.

Project MVP: all-transaction feature fusion



Feature fusion combines rule flags, anomaly scores and network features.

Every model scores all rows; time folds select XGBoost and validation sets its alert threshold.

Data, target and chronological split

SAML-D

9,504,852 transactions · 9,873 positive synthetic labels (0.104%)

Period	Dates	Rows	Positives	Positive rate
Training	Oct 7, 2022 to May 18, 2023	6,661,223	6,774	0.1017%
Validation	May 19 to Jul 5, 2023	1,430,805	1,416	0.0990%
Test	Jul 6 to Aug 23, 2023	1,412,824	1,683	0.1191%

Complete days stay in order. The test period is later than every training and threshold decision.

Features and leakage controls

Current transaction facts

Amount, payment type, currencies, bank locations, hour, weekday.

Past activity only

Sender, receiver and pair counts; earlier average amount; distinct counterparties.

Excluded from model inputs

Raw account IDs, timestamp, target label and generated laundering typology.

Temporal safeguard

Only events strictly before the row timestamp contribute to prior activity. Same-second rows do not see one another.

The typology field appears only in the post-score coverage audit.

Three detection tracks

Track	Signal for classifiers	Standalone alert rule	Test alerts
1 Rules	Three individual flags	Any of three rules fires	69,708
2 Anomaly	Broad Isolation Forest score	Behavior-focused IF; top 0.1% of validation scores	1,323
3 Network	Prior fan-in/out + degree score	Distinct relationships relative to pair repeats; top 0.1%	2,075

Important distinction

The alert-focused anomaly and network scores do not change the shared classifier inputs.

Model selection and thresholding

1	Expanding time folds	Mean AP compares ranking on later training windows
2	Full training fit	Each model refits on all training rows
3	Validation threshold	Each model chooses the threshold with highest F1
4	Later test	One held-out period supplies reported counts and rates

Why AP? The positive class is 0.104% of the dataset, so ranking quality and alert workload both matter.

XGBoost leads the later test comparison

Model	AP	Precision	Recall	F1	FPR	Alerts
Logistic regression	0.3176	44.2%	29.8%	0.3557	0.0449%	1,134
XGBoost	0.9900	96.6%	98.0%	0.9732	0.0041%	1,708
EBM	0.7815	81.5%	65.9%	0.7286	0.0179%	1,361

XGBoost at its validation threshold

1,650 true-positive labels · 58 false alerts · 33 positive labels missed

SAML-D XGBoost: earlier versus current

Same later test: 1,412,824 transactions · 1,683 positive labels

XGBoost run	AP	Precision	Recall	F1	FPR	Alerts	Caught	False	Missed
Earlier DS lane (Pattern 3-like)	0.8959	93.8%	84.1%	0.8872	0.0066%	1,509	1,416	93	267
Current MVP	0.9900	96.6%	98.0%	0.9732	0.0041%	1,708	1,650	58	33

+234
positive labels caught

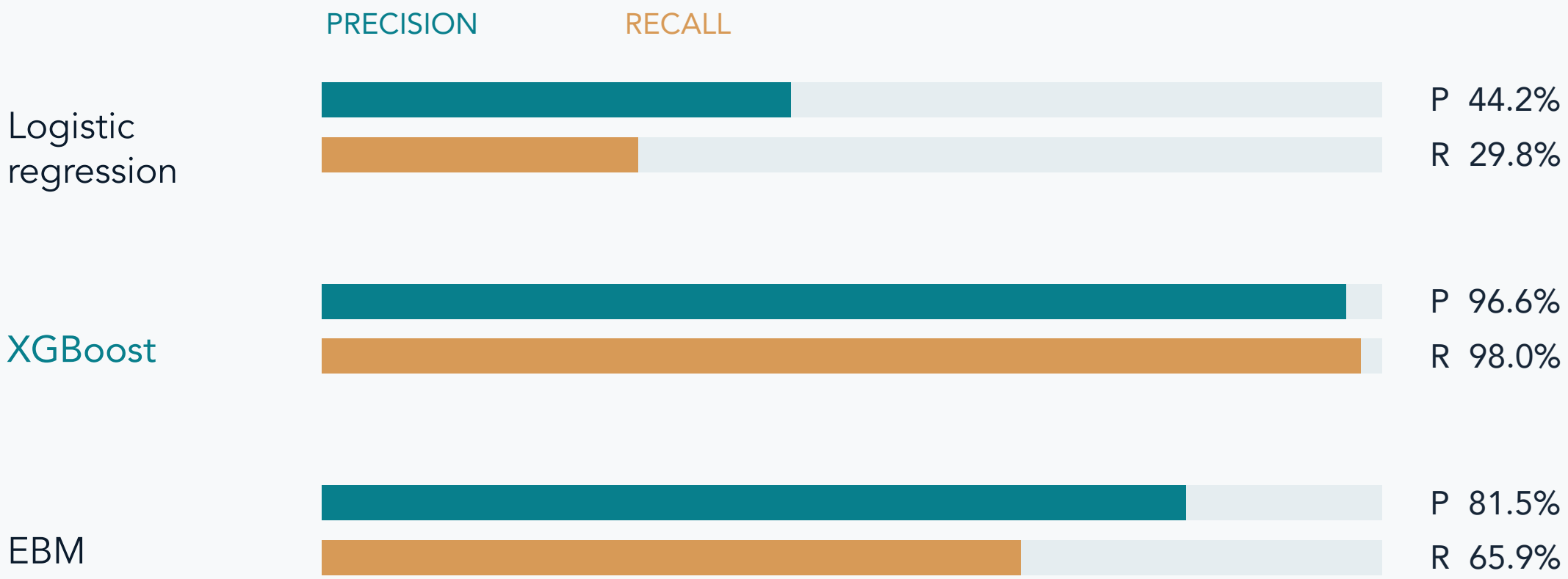
−234
positive labels missed

−35
false alerts

+199
total alerts

Pattern 3-like means the DS lane only. Historical runs changed features, so this is not a controlled ablation.

Precision and recall at the chosen thresholds



Precision divides by alerts. Recall divides by 1,683 positive test labels.

Standalone track alerts have different economics

Track	Alerts	Caught	False alerts	Precision	Recall
Fixed rules	69,708	648	69,060	0.93%	38.5%
Isolation Forest	1,323	45	1,278	3.40%	2.67%
Network fan-in/out	2,075	792	1,283	38.17%	47.06%

Track union 72,328 unique test alerts catch 1,196 positives, with 71,132 false alerts.

The union is a diagnostic benchmark, not the selected queue.

The union adds little coverage at very high cost

Selected XGBoost

1,708

alerts | 1,650 caught | 58 false

XGBoost plus all track alerts

72,795

alerts | 1,661 caught | 71,134 false

+11

additional positive labels

+71,076

additional false alerts

Isolation Forest and network alerts recover zero of XGBoost's 33 misses. Rules recover 11.

Explanations support review, not a verdict

Model	Explanation in the notebook	What it answers
Logistic regression	Coefficients and one-row contributions	Which scaled inputs move the log-odds
XGBoost	Global and local Tree SHAP	Which features move a tree score
EBM	Native global effects and local explanation	How learned feature effects add to a score

Interpretive limit

These plots explain model calculations. They do not prove laundering, validate synthetic labels or replace analyst judgment.

What a bank would validate before use

Outcome quality	Replace generator labels with documented case and investigation outcomes.
Coverage	Challenge performance by typology, customer segment and payment channel.
Operations	Set alert capacity, escalation policy and low-score sampling.
Governance	Independent validation, drift monitoring, change control and audit trail.

This class MVP has no customer context, analyst dispositions, case lifecycle or live monitoring controls.

Sources and project record

Project analysis	Notebook and class report in the linked repository
SAML-D	Author dataset repository and paper citation
FFIEC	BSA/AML Examination Manual, Suspicious Activity Reporting
FinCEN + agencies	Joint Statement on Innovative Efforts, 2018
FATF	Digital Transformation of AML/CFT
Federal Reserve	Current supervisory guidance on model risk management
BIS Innovation Hub	Project Aurora, network and ML research

Industry patterns are illustrative. All performance figures are from this synthetic class experiment.

REFERENCES